

AI HEALTH SYMPTOM CHECKER REPORT

1. ABSTRACT

This project focuses on developing an AI Health Symptom Checker using Python, Streamlit, SQLite, and Machine Learning techniques. The system allows users to enter symptoms and receive AI-based disease predictions along with risk analysis and health recommendations.

The main objective of this project is to provide a smart healthcare assistant that helps users understand possible health conditions based on symptom severity. The system also stores prediction history and provides analytics through an admin dashboard.

This project demonstrates how artificial intelligence and data visualization can improve healthcare accessibility and early symptom analysis.

2. INTRODUCTION

Healthcare systems generate large amounts of patient and symptom-related data. Early disease prediction and symptom analysis can help users take preventive actions before conditions become severe.

This project aims to develop an AI-powered Health Symptom Checker using Streamlit. Users can register, log in, enter symptom severity, and receive disease predictions generated using a Random Forest Machine Learning model.

The application also includes BMI analysis, risk assessment, multilingual support, history tracking, and admin analytics dashboards for better user experience.

3. OBJECTIVES

- To develop an AI-based disease prediction system
 - To analyze symptom severity
 - To calculate health risk levels
 - To provide BMI analysis and recommendations
 - To maintain user prediction history
 - To create an interactive analytics dashboard
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4. SYSTEM IMPLEMENTATION

The system is implemented using Python, Streamlit, SQLite, and Scikit-learn with the following components:

4.1 Database Management

- SQLite database (**health_app.db**)
- Stores user login and prediction history

4.2 User Authentication

- Signup and login functionality
- Password hashing using SHA-256

4.3 Symptom Input System

- Users provide symptom severity using sliders
- Symptoms include fever, cough, headache, fatigue, etc.

4.4 Machine Learning Model

- Random Forest Classifier for disease prediction
- Predicts diseases such as:
 - COVID-19
 - Flu
 - Cold
 - Viral Fever

4.5 Risk Analysis

- Calculates risk level (Low, Medium, High)
- Based on total symptom severity score

4.6 BMI Analysis

- Calculates BMI using height and weight
- Displays health status:
 - Underweight
 - Normal
 - Overweight

- Obese

4.7 Data Visualization

- Symptom severity analysis chart
- Disease probability pie chart
- Admin analytics dashboard

4.8 AI Recommendations

- Provides health suggestions based on risk level
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5. FUNCTIONAL MODULES

5.1 Signup/Login

User authentication system

5.2 Health Checker

Disease prediction using AI model

5.3 BMI Calculator

Calculates body mass index

5.4 Prediction History

Stores and displays previous reports

5.5 Admin Dashboard

Displays disease and risk analytics

5.6 Multi-language Support

Supports:

- English
 - Gujarati
 - Hindi
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6. WORKING FLOW

1. User signs up or logs in
2. User enters health details and symptoms

3. System calculates BMI and symptom score
 4. Machine learning model predicts disease
 5. Risk level and confidence score are displayed
 6. Results are stored in database
 7. Admin dashboard displays analytics
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7. KEY FEATURES

- AI-based disease prediction
 - Interactive Streamlit interface
 - Symptom severity analysis
 - BMI calculator
 - Prediction confidence score
 - Multi-language support
 - Admin analytics dashboard
 - Health recommendation system
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8. ADVANTAGES

- Easy to use interface
 - Provides instant health insights
 - Supports multiple languages
 - Tracks prediction history
 - Uses machine learning for analysis
 - Interactive charts and analytics
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9. LIMITATIONS

- Educational purpose only
- Limited disease dataset
- No real-time medical integration

- Basic machine learning model
 - Not a replacement for professional healthcare
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10. FUTURE ENHANCEMENTS

- Integrate real medical datasets
 - Add advanced AI/Deep Learning models
 - Implement doctor consultation system
 - Add cloud database integration
 - Develop mobile application
 - Add voice assistant support
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11. CONCLUSION

The AI Health Symptom Checker successfully demonstrates how Python, Streamlit, and machine learning can be used to build an intelligent healthcare assistant.

The system provides disease prediction, BMI analysis, risk assessment, and interactive visualizations, helping users understand their health conditions better. With further enhancements, it can evolve into a more advanced healthcare support platform.